The Impact of Genomic Changes on Treatment of Lung Cancer

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The remarkable success of epidermal growth factor receptor (EGFR) and anaplastic lymphoma kinase (ALK) tyrosine kinase inhibitors in patients with *EGFR* mutations and *ALK* rearrangements, respectively, introduced the era of targeted therapy in advanced non-small cell lung cancer (NSCLC), shifting treatment from platinum-based combination chemotherapy to molecularly tailored therapy. Recent genomic studies in lung adenocarcinoma identified other potential therapeutic targets, including *ROS1* rearrangements, *RET* fusions, *MET* amplification, and activating mutations in *BRAF*, *HER2*, and *KRAS* in frequencies exceeding 1%. Lung cancers that harbor these genomic changes can potentially be targeted with agents approved for other indications or under clinical development. The need to generate increasing amounts of genomic information should prompt health-care providers to be mindful of the amounts of tissue needed for these assays when planning diagnostic procedures. In this review, we summarize oncogenic drivers in NSCLC that can be currently detected, highlight their potential therapeutic implications, and discuss practical considerations for successful application of tumor genotyping in clinical decision making.

Keywords: lung cancer; cancer genomics; molecular targeted therapy

Lung cancer remains the leading cause of cancer-related mortality in the United States and worldwide. Approximately 85% of patients are diagnosed with non-small cell lung cancer (NSCLC), and most present with advanced disease that is not amenable to curative therapy. For these patients, cytotoxic chemotherapy offers modest prolongation in survival. Over the past 2 decades, modifications of chemotherapy combinations, the addition of monoclonal antibodies, including bevacizumab and cetuximab, and the incorporation of histologic subtype into treatment decisions have added incrementally to the survival of patients with advanced NSCLC. However, therapeutic outcomes appear to have reached a plateau, with response rates of 20 to 35% and median survival of 8 to 12 months (1–3). Recently, clinical and laboratory research have led to a better understanding of the molecular pathogenesis that causes lung cancer. The ongoing research of patients with NSCLC and their tumors has demonstrated that NSCLC can be further defined at the molecular level by the identification and characterization of oncogenic drivers that occur in genes critical to cellular proliferation and survival, leading to aberrant activation of signaling proteins that induce sustained growth of the lung cancer cells. These advances in the molecular pathogenesis of NSCLC have prompted investigators to systematically characterize lung cancers for these oncogenic drivers and treat with appropriate targeted therapies. A cohort of patients can now be prospectively identified who have significantly improved outcomes with such therapies, with median survival in excess of 2 years, necessitating the routine identification of these genomically defined patient subsets. In this perspective, we review oncogenic drivers in NSCLC that can be currently detected and their potential impact on treatment now and in the future (Table 1). This review focuses on driver events that have been identified in lung adenocarcinomas, as this is where the field is most mature from a therapeutic perspective. Recent studies have also uncovered potentially actionable genetic changes in squamous cell lung cancers (Figure 1) (4, 5). To highlight therapeutic implications, this perspective discusses oncogenic drivers grouped by the availability of targeted agents for the specific genetic alteration. Some of the results of the studies discussed in this review have been previously reported in the form of an abstract (6–11).

ONCOGENIC DRIVERS WITH APPROVED TARGETED AGENTS

Epidermal Growth Factor Receptor Mutations

The epidermal growth factor receptor (EGFR) is a transmembrane signaling receptor that plays a central role in several cellular processes, including proliferation, migration, adhesion, and invasion. The EGFR was proposed as a therapeutic target based on its overexpression in multiple epithelial cancers, including NSCLC. Initial clinical trials of the oral EGFR tyrosine kinase inhibitors (TKIs) erlotinib and gefitinib showed modest efficacy after chemotherapy in unselected patients with NSCLC, with a response rate of about 10% and a 2-month prolongation in median survival over placebo (12), and noninferiority compared with docetaxel (13), respectively. A key early observation with gefitinib and erlotinib was that the rare responses were often dramatic and durable for a year or longer, an unusual finding for chemotherapy-treated patients. In 2004, investigators from our center and others discovered an underlying association between activating mutations within the tyrosine kinase (TK) domain of *EGFR* and EGFR TKI–responsive patients with NSCLC (14, 15). These *EGFR* mutations, most frequently exon 19 deletions

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TABLE 1. ONCOGENIC DRIVERS IN LUNG ADENOCARCINOMAS

Oncogenic Driver	Prevalence, %*	Potential Available or Investigational Targeted Agent(s)
Oncogenic drivers with approved targeted agents		
EGFR mutations	10–15	Erlotinib[†] Gefitinib[†] Afatinib
ALK rearrangements	3–7	Crizotinib[†]
Oncogenic drivers with approved targeted agents in other tumor types		
ROS1 rearrangements	1–2	Crizotinib[†]
BRAF mutations	2–4	Vemurafenib[‡] Dabrafenib[‡] Trametinib[‡] Dasatinib[‡]
HER2 insertions	2	Afatinib Neratinib Dacomitinib
MET amplification[§]	<1	Crizotinib[†]
RET fusions	1–2	Cabozantinib[‡] Vandetanib[‡] Sorafenib[‡] Sunitinib[‡]
Potentially targetable oncogenic drivers with investigational agents		
KRAS mutations	20	Selumetinib

Definition of abbreviations: *ALK* = anaplastic lymphoma kinase; *EGFR* = epidermal growth factor receptor.

*In most cases, the oncogenic drivers identified in lung adenocarcinomas are mutually exclusive from one another.

[†] Agent approved by the U.S. Food and Drug Administration in lung cancer.

[‡] Agent approved by the U.S. Food and Drug Administration in another tumor type.

[§] Primary (*de novo*) *MET* amplification.

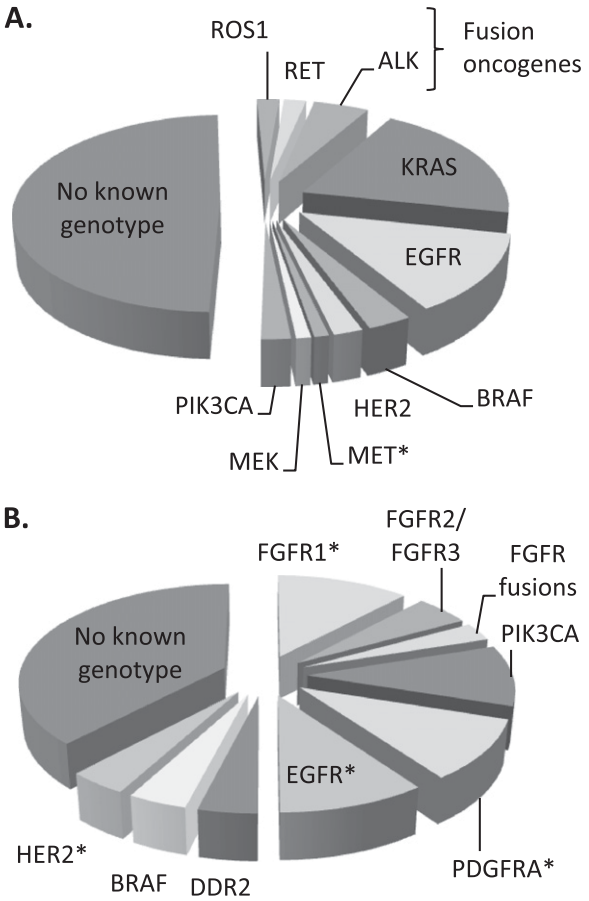

Figure 1. Targetable and/or potentially targetable genetic changes in (*A*) adenocarcinoma and (*B*) squamous cell carcinomas of the lung. *Gene amplification. ALK = anaplastic lymphoma kinase; EGFR = epidermal growth factor receptor.

or the exon 21 point mutation L858R, cause ligand-independent activation of EGFR signaling and simultaneously enhance sensitivity to EGFR TKI therapy. Mutations in *EGFR* are present in 10 to 15% of patients with NSCLC from a European background and are more common in never smokers, women, Asians, and patients with adenocarcinoma.

The initial observations of the association between gain-of-function mutations of *EGFR* and sensitivity to EGFR TKIs led to a series of trials comparing the outcomes of patients with *EGFR* mutations treated with EGFR TKIs versus those treated with chemotherapy. Seven studies were undertaken that enrolled patients either selected on the basis of clinical features commonly associated with *EGFR* mutations or prospectively identified with sensitizing mutations of *EGFR* and randomized to EGFR TKI or combination chemotherapy. These seven randomized studies comparing first-line erlotinib, gefitinib, or the second-generation EGFR TKI, afatinib, to chemotherapy have uniformly demonstrated the superiority of EGFR TKIs in terms of progression-free survival (PFS), response, tolerability, and quality of life (16). In contrast, an overall survival (OS) benefit from EGFR TKI therapy in the *EGFR* mutation–positive population has not yet been demonstrated, likely because of crossover rates of 59 to 98% from chemotherapy to EGFR TKI at disease progression reported in the trials. When given as first-line therapy for *EGFR*-mutated disease, gefitinib, erlotinib, and afatinib yield responses in 60 to 80% of patients, median PFS of 9 to 12 months, and median survival in excess of 2 years from the start of EGFR TKI therapy. Mutations in *EGFR* are also a potential positive predictive marker for efficacy of platinum-based combination chemotherapy in advanced NSCLC (17). Current National Comprehensive Cancer Network (NCCN), American Society of Clinical Oncology (ASCO), European Society for Medical Oncology (ESMO), and College of American Pathologists (CAP)/International Association for the Study of Lung Cancer (IASLC)/Association for Molecular Pathology (AMP) guidelines agree on the importance of early characterization of *EGFR* in the treatment algorithm for advanced nonsquamous NSCLC. Patients with *EGFR*-mutant advanced NSCLC should be offered treatment with an EGFR TKI during the course of their disease, preferably in the first-line setting, as this strategy improves PFS, treatment-adverse effect profiles, and quality of life. Gefitinib and erlotinib have been approved in Europe and in the United States, respectively, as treatment for *EGFR*-mutant NSCLC. Approval of afatinib is currently under review by the U.S. Food and Drug Administration (FDA) and European Medical Agency.

Despite the initial activity of gefitinib and erlotinib in the treatment of *EGFR*-mutant NSCLC, all EGFR TKI–responsive patients eventually develop resistance to the EGFR TKI. Strategies aimed at overcoming and/or preventing acquired resistance to EGFR TKIs have been recently reviewed elsewhere (18).

Anaplastic Lymphoma Kinase Rearrangements

The second biomarker linked to an approved use of a targeted agent in patients with advanced NSCLC is a rearrangement of the anaplastic lymphoma kinase (*ALK*) gene. In 2007, investigators from Japan identified a fusion gene arising from an intrachromosomal

translocation in lung adenocarcinomas, which causes a rearrangement between the *ALK* gene and an upstream partner, *EML4* (19). The resulting EML4-ALK fusion protein possesses potent oncogenic activity. *EML4-ALK* is the predominant *ALK* fusion in NSCLC, although other *ALK* fusions have been reported. Further studies have demonstrated that rearrangements of the *ALK* gene are present in 3 to 7% of unselected patients with NSCLC (19, 20). Similar to *EGFR* mutations, *ALK* rearrangements are associated with distinct clinical features, including younger age at NSCLC diagnosis, absent or minimal smoking history, and adenocarcinoma histology.

Crizotinib is a small molecule inhibitor most potent against MET but also active against other receptor tyrosine kinases, including ALK and ROS1. Phase I and II studies of crizotinib in previously treated patients with *ALK*-rearranged advanced NSCLC showed response rates in excess of 60% and PFS greater than 8 months (7, 21). By comparison, standard single-agent chemotherapy in previously treated advanced NSCLC yields response rates of 10% or less, and median PFS of 2 to 3 months. Crizotinib-related toxicities were predominantly grade 1 or 2, with rare grade 3 or 4 toxicities that included pneumonitis (1.6% across phase I and II studies). The striking results of the phase I and II studies, along with the favorable safety profile of crizotinib, led to the 2011 accelerated FDA approval of crizotinib for treatment of *ALK*-positive advanced NSCLC as defined by a jointly approved diagnostic test using a break-apart fluorescence *in situ* hybridization (FISH) assay. Since then, results of the randomized phase III trial comparing second-line crizotinib with pemetrexed or docetaxel in patients with advanced *ALK*-positive NSCLC have demonstrated significantly higher response rates (65 vs. 20%; $P < 0.0001$) and longer PFS with crizotinib (median PFS, 7.7 vs. 3.0 mo; hazard ratio (HR), 0.49; 95% confidence interval, 0.37–0.64) (22). Results of the first-line randomized phase III trial (PROFILE 1014) comparing crizotinib with a platinum-pemetrexed combination in patients newly diagnosed with advanced *ALK*-positive NSCLC are awaited, with PFS as the primary endpoint. OS is a secondary endpoint and likely to be confounded by crossover. A recent retrospective analysis comparing 82 patients with *ALK*-positive NSCLC treated with crizotinib in the phase I trial with 36 similar patients who never received the drug suggested a significant prolongation in survival with targeted therapy, with a median survival in excess of 2 years from the start of crizotinib therapy (23). Current guidelines from the NCCN and CAP/IASLC/AMP recommend *ALK* testing of advanced nonsquamous NSCLC and crizotinib as initial therapy for patients with *ALK*-positive NSCLC.

Similar to EGFR TKIs, acquired resistance ultimately limits the clinical benefit of crizotinib. Clinical trials of next-generation ALK inhibitors are ongoing, and strategies aimed at overcoming and/or preventing the emergence of resistance in *ALK*-positive NSCLC are being explored.

ONCOGENIC DRIVERS WITH APPROVED TARGETED AGENTS IN OTHER TUMOR TYPES

ROS1 Rearrangements

ROS1 is a receptor tyrosine kinase with homology to the insulin receptor. Chromosomal rearrangements involving the *ROS1* gene were originally described in glioblastomas before their identification in NSCLC in 2007 (24). Several *ROS1* fusion partners have been reported in NSCLC. These *ROS1* rearrangements lead to constitutive kinase activity. Prevalence screens using a break-apart FISH assay have demonstrated that approximately 1 to 2% of lung adenocarcinomas harbor *ROS1* rearrangements (25, 26). In one series, *ROS1* rearrangements were found to be more common in never or light smokers and were associated with younger age at diagnosis, a clinical profile similar to that of patients with *ALK*-rearranged NSCLC (25).

In vitro studies demonstrated that the ALK/MET inhibitor, crizotinib, inhibits the growth of *ROS1*-positive lung cancers at achievable serum concentrations (25). Preliminary efficacy and safety data on the first 15 *ROS1*-positive patients enrolled onto an expansion cohort of the ongoing phase I study PROFILE 1001 of crizotinib were recently presented (10). Among 14 evaluable patients, the response rate was 57%, a result remarkably similar to that seen with *ALK*-positive patients treated with crizotinib. *ROS1*-specific kinase inhibitors are also in clinical development. The identification of *ROS1*-rearranged NSCLC is expected to build on the ALK model for rapid validation of a biomarker that will likely have an impact on diagnosis and treatment of lung cancer. The NCCN guidelines currently suggest crizotinib therapy in patients with advanced NSCLC bearing a *ROS1* rearrangement.

BRAF Mutations

BRAF is a serine/threonine kinase that lies downstream of RAS in the RAS/RAF/MEK/ERK signaling pathway. Mutations in *BRAF* are seen in approximately one-half of melanomas, in which *BRAF* V600E is a driver mutation that can be effectively targeted with selective BRAF and/or MEK inhibitors. Mutations in *BRAF* are also detected in 2 to 4% of NSCLCs (27–29). The mutations found in NSCLC are distinct from the melanoma setting: whereas *BRAF*-mutated melanomas harbor a V600E amino acid substitution in more than 80% of cases, lung adenocarcinomas harbor non-V600E mutations in 40 to 50% of cases. Many non-V600E mutations show only intermediate or low kinase activity, and the classification of some as driver events remains less certain than other oncogenic changes. Two recent studies suggested that non-V600E *BRAF* mutations occur almost exclusively in smokers (28, 29); one group found that *BRAF* V600E was more common in never smokers and in women (28).

Because of the predominance of *BRAF* V600E mutations in melanoma, drugs targeting BRAF, including vemurafenib and dabrafenib, are designed to have specific activity against the V600E mutant BRAF kinase. Preliminary results from an ongoing phase II trial of dabrafenib described eight responses out of 20 evaluable patients with *BRAF* V600E mutant NSCLC (NCT01336634) (9). However, preclinical data suggest that non-V600E mutant BRAF kinases are resistant to BRAF inhibitors, although some may be sensitive to downstream pathway inhibitors. Agents targeting BRAF or downstream effectors in ongoing clinical trials include the BRAF inhibitor, dabrafenib, for patients with NSCLC and prospectively identified *BRAF* V600E mutations; the MEK inhibitor, trametinib, for patients with non-V600E *BRAF* mutations; and dasatinib for patients with NSCLC and inactivating or uncharacterized *BRAF* mutations (NCT01336634, NCT01362296, and NCT01514864, respectively).

HER2 Amplification and Insertions

HER2 is a member of the HER family of receptors that is activated in 25 to 30% of breast cancers by focal genomic amplification. *HER2*-amplified breast cancer can be effectively treated with the anti-HER2 monoclonal antibody, trastuzumab, and/or the HER2 TKI, lapatinib. HER2 receptor expression is detectable by immunohistochemistry in approximately 20% of NSCLCs at levels lower than those observed in HER2-positive breast cancer. The trials of trastuzumab in HER2-expressing NSCLC using a lower criterion than that for breast cancer failed to demonstrate prolonged efficacy or survival when administered as

monotherapy or in combination with chemotherapy (30, 31). *HER2* gene amplification detected by FISH using criteria for amplification in breast cancer is present in approximately 2% of NSCLCs. Data are incomplete regarding the efficacy of trastuzumab or other HER2-targeted agents in a population with this more rigorous definition of *HER2* amplification (32).

HER2 is also activated by exon 20 in-frame insertion mutations in approximately 2% of lung adenocarcinomas; these mutations are not seen in breast cancer (33, 34). Several TKIs are being tested in *HER2*-dependent lung adenocarcinomas. A phase II trial of the EGFR/HER2 dual inhibitor afatinib showed responses in three of five pretreated patients with advanced NSCLC with prospectively identified *HER2* mutations (35). The preliminary results from an ongoing phase II trial of dacomitinib, another oral pan-HER irreversible TKI, described three responses out of 18 evaluable patients with *HER2*-mutant NSCLC (NCT00818441) (8). Promising results were also reported from a phase I trial of the irreversible pan-HER inhibitor, neratinib, and the mTOR inhibitor, temsirolimus, with partial responses in two of six patients with *HER2*-mutant NSCLC (6). A phase II trial of neratinib versus neratinib plus temsirolimus in patients with *HER2*-mutant NSCLC is planned (NCT01827267).

MET Amplification

The *MET* protooncogene encodes a transmembrane tyrosine kinase receptor that activates downstream signaling molecules involved in cell proliferation, survival, motility, and invasion. Amplification of the *MET* gene has been associated with the development of resistance to gefitinib or erlotinib in approximately 5% of patients with sensitizing *EGFR* mutations who were initially responsive to the drug (36, 37). Primary (*de novo*) *MET* gene amplification detected by FISH using criteria for enrollment in the *MET*-enriched cohort of the ongoing phase I study of crizotinib (*MET*/*CEP7* ratio > 2.2; not polysomy) has been identified in less than 1% of patients with lung adenocarcinomas not previously treated with EGFR TKIs and in approximately 6% of squamous cell lung carcinomas (38, 39). Preclinical studies have shown that *MET* amplification has transforming activity. Somatic mutations in the TK domain of *MET* have also been described in NSCLC, but their biologic significance remains incompletely characterized (40).

The approved ALK inhibitor, crizotinib, was initially developed as a potent inhibitor of MET. There is one report in the literature of rapid and durable clinical response in a patient with NSCLC with *de novo MET* amplification and no rearrangement of the *ALK* gene enrolled in the MET-enriched cohort of the ongoing phase I study of crizotinib (PROFILE 1001), suggesting that primary *MET* amplification may be an oncogenic driver in a subset of NSCLCs and a potential therapeutic target (41). Other small molecule inhibitors are in clinical trial to assess their ability to target the MET pathway. Antibodies directed against the MET receptor or its ligand, hepatocyte growth factor, are also being evaluated in clinical trials. Efficacy data were recently presented for MetMAb, an anti-MET receptor monoclonal antibody, from a randomized double-blind phase II study comparing MetMAb plus erlotinib with placebo plus erlotinib in patients with previously treated NSCLC who were not selected by genotype. In a planned subgroup analysis of MET-diagnostic–positive patients determined by immunohistochemistry, MetMAb plus erlotinib significantly prolonged PFS (2.9 vs. 1.5 mo; HR, 0.53; $P = 0.04$) and OS (12.6 vs. 3.8 mo; HR, 0.37; $P = 0.002$) (11). A phase III trial is underway in previously treated patients with NSCLC prospectively identified with MET overexpression (NCT01456325).

RET Fusions

The *RET* protooncogene encodes a TK receptor that is involved in cell proliferation, migration, differentiation, and neuronal navigation. Germline and somatic gain-of-function *RET* mutations are known to predispose to multiple endocrine neoplasia type 2 and sporadic medullary thyroid cancer, respectively, whereas somatic *RET* gene fusions account for the majority of radiation-induced and sporadic papillary thyroid cancers. In 2011, investigators from Seoul identified a fusion gene involving *RET* partnered with *KIF5B* in a young never smoker with lung adenocarcinoma (42). *RET* rearrangements have since been identified in approximately 1 to 2% of patients with adenocarcinoma or adenosquamous carcinoma of the lung. *KIF5B-RET* is the predominant *RET* fusion in NSCLC, although other *RET* fusions have been reported (26, 43, 44). In a series of 13 cases, patients with *RET*-positive NSCLC tended to be younger and never smokers, sharing similar features with *ALK*- and *ROS1*-positive patients (45).

KIF5B-RET and other *RET* fusions induce constitutive activation of the oncoprotein. Cells expressing *KIF5B-RET* are sensitive *in vitro* to multitargeted TKIs that inhibit RET, such as vandetanib, sorafenib, and sunitinib (26, 43, 44). Cabozantinib, a multi-TKI and more potent inhibitor of RET, is approved by the FDA for the treatment of advanced medullary thyroid cancer and is currently being tested in a phase II trial for *RET* fusion–positive NSCLC (NCT01639508). Preliminary efficacy data on the first three patients enrolled on this study demonstrated partial responses in two patients and prolonged disease stabilization in the third patient (46). This report provides early clinical validation of oncogene addiction in *RET*-positive lung cancers and suggests that *RET* fusions may expand the repertoire of somatic alterations for which targeted therapy may be offered in patients with advanced NSCLC.

POTENTIALLY TARGETABLE ONCOGENIC DRIVERS WITH INVESTIGATIONAL AGENTS

KRAS Mutations

KRAS is the *RAS* family gene most frequently activated in lung cancers, with mutations detected in about 20% of NSCLCs, most commonly in adenocarcinomas. Most *KRAS* mutations are single amino acid substitutions in codons 12, 13, or 61. These mutations render KRAS constitutively active, leading to activation of downstream effectors, including the RAF/MEK/ERK and PI3K/AKT/mTOR signaling cascades. *KRAS* mutations are associated with a history of tobacco use; current or former smokers most commonly have transversion mutations, whereas never smokers are more likely to harbor transition mutations. There are incomplete data regarding the functional consequence to the different types of *KRAS* mutations in NSCLC.

Mutations in *KRAS* are a potential negative predictive marker for efficacy of EGFR TKIs in patients with NSCLC with wild-type *EGFR* as well as a potential therapeutic target. Results were recently reported from a randomized phase II study comparing selumetinib, an oral MEK1/MEK2 inhibitor, in combination with docetaxel with docetaxel plus placebo in the second-line treatment of 87 patients with advanced NSCLC bearing *KRAS* mutations (47). A numerically longer OS was observed in the combination arm, but this did not reach statistical significance, possibly due to low power because of small sample sizes. However, selumetinib and docetaxel compared with docetaxel and placebo significantly prolonged PFS (5.3 vs. 2.1 mo; HR, 0.58; 80% confidence interval, 0.42–0.79). A larger phase III trial is needed as a confirmatory study. Other agents targeting KRAS or downstream pathways in ongoing clinical trials include the

MEK inhibitor, trametinib; the MET inhibitor, tivantinib, plus erlotinib; or the hsp90 inhibitor, IPI-504, plus the mTOR inhibitor, everolimus, for patients with NSCLC and prospectively identified *KRAS* mutations (NCT01362296, NCT01244191, and NCT01427946, respectively).

CLINICAL APPLICATION FOR DIAGNOSTIC SPECIMENS

The advent of effective, targeted therapies for molecularly defined subsets of patients with NSCLC has prompted the need for more extensive genomic characterization of advanced NSCLC to guide the selection of first-line systemic therapy and to identify potential candidates for investigational therapy. Such personalized lung cancer therapy has raised practical challenges. Foremost, both quantity and quality of tumor tissue are critical for successful genomic testing, stressing the need to obtain adequate tumor material at the time of diagnostic sampling. Because most patients with lung cancer present with advanced-stage disease, the information to guide treatment algorithms frequently relies on small biopsy or cytology specimens, such as pleural fluid or fine-needle aspirates. Biopsies with small (18- to 20-gauge) core needles can yield sufficient and reliable samples for genomic testing (48). Similarly, groups have been optimizing techniques for nonsurgical sampling, such as endobronchial ultrasound-guided transbronchial needle aspiration, and mutation assays have been shown to be able to detect somatic mutations from cytology specimens, particularly if cell blocks are available (49, 50). Ultimately, close communication between the clinical care team, testing laboratory, and pathologist is needed to ensure that specimens for molecular testing meet the laboratory's requirements for tumor content and quality. Another practical challenge is the length of time needed to perform genomic testing, which must be short to successfully integrate tumor molecular analysis into the fast pace of clinical decision making. Guidelines by the CAP/IASLC/AMP recommend that *EGFR* and *ALK* testing both be completed within 10 working days of receiving the specimen in the testing laboratory (49). Moving forward, the adaption of new technologies, such as next-generation sequencing, may offer the possibility of rapidly and comprehensively interrogating the cancer genome of individual patients, thereby facilitating the identification of actionable and/or novel oncogenic alterations and targeted treatment options.

Author disclosures are available with the text of this article at www.atsjournals.org.

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